Jianjian Gao

Postdoc research associate

School of Data Science, University of Virginia, Charlottesville, Virginia, USA

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https://jiaoyang2018.github.io/

Research Interests

Computational social science; Science of science; Causal inference; Network science

I mainly conduct quantitative research in the field of Science of Science and Computational Social Science using various quantitative methods, including network analysis, causal inference and machine learning.

Academic Experience

2023-present **Postdoc research associate**, School of Data Science, University of Virginia.
Co-advised by: Professor Yong-Yeol Ahn (yyahn@virginia.edu); Assistant Professor Alexander J. Gates (hgt6rn@virginia.edu)

2017-2018 **Research assistant**, Department of Energy and Power Engineering, Tsinghua University.

Education

Queen Mary University of London *London*

PhD in Business and Management (Computational Social Science) *2018 - 2022*

- Supervisor: Professor Pietro Panzarasa (p.panzarasa@qmul.ac.uk)

Institute of Engineering Thermophysics, Chinese Academy of Sciences *Beijing*

Master in Power Engineering *2013 - 2016*

- Supervisor: Professor Hui Hong; GPA: 84.41/100

Shandong University *Shandong*

Bachelor in Thermal Energy and Power Engineering *2009 - 2013*

- GPA: 87.91/100. Outstanding Graduates Awards of Shandong Province (2013); National Endeavour Scholarship (2012); Research and Innovation of Advanced Personal (2011); The First Prize Scholarship (2010); Merit student (2010)

Industry Experience

2016-2017 **Assistant engineer**, Beijing Petrochemical Engineering Co., LTD.(BPEC)

Publications

Under Review

Alexander J. Gates, **Jianjian Gao**, Indraneel Mane. The increasing fragmentation of global science and the diffusion of ideas. Nature Communications. https://arxiv.org/abs/2404.05861

Jianjian Gao, Alexander J. Gates. Geographic inequality in contemporary art exhibitions arises from nationality bias and unequal returns to places. Recently submitted. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6907438

Working papers

Ty Benjamin Misiorek, **Jianjian Gao**, Alexander J. Gates. The diversity of philanthropic boards and the diffusion of scientific policy.

Carattini Stefano, Sam Fankhauser, **Jianjian Gao**, Caterina Gennaioli, and Pietro Panzarasa. Economic integration and climate change laws.

Jianjian Gao, Pietro Panzarasa. Regionaliszation of international environmental cooperation: Evidence from community structure analysis.

Published

Jianjian Gao, Alexander J. Gates. Knowledge Sovereignty: A Path to Independence or Isolation? Research Policy. 2026, 55(8):105557 (https://doi.org/10.1016/j.respol.2026.105557)

Carattini Stefano, Sam Fankhauser, **Jianjian Gao**, Caterina Gennaioli, and Pietro Panzarasa. What does network analysis teach us about international environmental cooperation? Ecological Economics. 2023, 205:107670. (**One chapter of the PhD thesis**)

Stefano Carattini, Sam Fankhauser, **Jianjian Gao**, Caterina Gennaioli, Pietro Panzarasa. The global network of environmental agreements: a preliminary analysis. Annual Bank Conference on Development Economics 2019, Washington, DC.

Jianjian Gao. International environmental cooperation and climate change laws: A quantitative analysis. 2022. Queen Mary University of London (**PhD thesis**).

Hui Hong, **Jianjian Gao**, Wanjun Qu, et al. Thermodynamic analyses of the solar-driven Kalina cycle having a variable concentration ratio. Applied Thermal Engineering. 2017, 126: 997-1005. (**Part of Master thesis**)

Jianjian Gao, Hui Hong, Jie Sun, et al. Study on the performance of concentrated solar Kalina cycle with variable mirror area parabolic trough collector. Journal of Engineering Thermophysics. 2016, 37(8):1595-1601. (EI)

Hao Zhang, Hui Hong, **Jianjian Gao**, et al. Thermodynamic performance of a mid-temperature solar fuel system for cooling, heating and power generation. Applied Thermal Engineering, 2016, 106:1268-1281.

Patents

Hui Hong, Jie Sun, **Jianjian Gao**. Wide-range irradiation and without energy storage concentrating solar energy Kalina power generation system and method. Beijing: CN105156285A, 2015.12.16.

Presentations

NetSci: International School & Conference on Network Science; **IC2S2**: International Conference on Computational Social Science; **ICSSI**: International Conference on the Science of Science; **Sunbelt**: Sunbelt Conference of the International Network for Social Network Analysis (INSNA); **ICA**: Annual International Communication Association Conference; **ISSI**: International Society for Scientometrics and Informetrics.

ICSSI, June 29-July 1, 2026, Denver, US – **Jianjian Gao**, Alexander J. Gates. The geography of scientific recognition across disciplines. (Talk)

ICSSI, June 29-July 1, 2026, Denver, US – **Jianjian Gao**, Alexander J. Gates. Knowledge Sovereignty: A Path to Independence or Isolation? (Poster)

NetSci, June 1-5, 2026, Boston, US – **Jianjian Gao**, Alexander J. Gates. Geographic inequality in contemporary art exhibitions arises from nationality bias and unequal returns to places. (Poster)

NetSci, June 1-5, 2026, Boston, US – **Jianjian Gao**, Alexander J. Gates. The geography of scientific recognition across disciplines. (Talk)

Sunbelt, June 23-29, 2025, Paris, France – Ty Benjamin Misiorek, **Jianjian Gao**, Alexander J. Gates. Connected boards, diverse leaders: How networks shape university presidencies. (Talk)

ISSI, June 23-27, 2025, Yerevan, Armenia. **Jianjian Gao**, Alexander J. Gates. Country self-preference and national research systems: A path to independence or isolation? (Talk)

ICA, June 12-16, 2025, Denver, Colorado, USA – Gates, Alexander J., **Jianjian Gao**, Indraneel Mane. The increasing fragmentation of global science limits the diffusion of ideas. (Lightning talk)

ICSSI, Jul 1-3, 2024, DC, USA – **Jianjian Gao** and Gates, Alexander J. The changing landscape of academic leadership: A study of presidents at US R1s. (Poster)

ICSSI, Jul 1-3, 2024, DC, USA – Gates, Alexander J., Indraneel Mane, and **Jianjian Gao**. The increasing fragmentation of global science limits the diffusion of ideas. (Lightning talk)

IC2S2, Jul 18-20, 2024, Philadelphia, USA – Gates, Alexander J., Indraneel Mane, and **Jianjian Gao**. The increasing fragmentation of global science limits the diffusion of ideas. (Poster)

NetSci, Sep 17-25, 2020, Roma, Italy – **Jianjian Gao**, Caterina Gennaioli, Pietro Panzarasa. Communities and rich clubs in the international environmental cooperation network. International School and Conference on Network Science. (Poster)

IC2S2, 2020, MA USA – **Jianjian Gao**, Stefano Carattini, Sam Fankhauser, Caterina Gennaioli, Pietro Panzarasa. Structure and Evolution of the International Environmental Cooperation Network. 6th

International Conference on Computational Social Science. (Talk)

IC2S2, 2020, MA USA – **Jianjian Gao**, Caterina Gennaioli, Pietro Panzarasa. Communities and rich clubs in the international environmental cooperation network. 6th International Conference on Computational Social Science. (Poster)

IC2S2, 2019, Netherlands – **Jianjian Gao**, Caterina Gennaioli, Pietro Panzarasa. Structure and evolution of the network of countries signing global environmental treaties. 5th International Conference on Computational Social Science. (Poster)

Teaching Experience

Queen Mary University of London *London*

Teaching Assistant *Spring 2022*

- **Experiments for business and analytics (BUSM106)**.
- This course deals with **advanced econometrics** and **R** programming for Master's students. I mainly taught students how to use **R** to perform all kinds of causal inference, including Difference in Difference (DID), Instrumental Variables (IV), Regression Discontinuity (RD), matching, etc.

Queen Mary University of London *London*

Teaching Assistant *Fall 2022*

- **Quantitative Research Methods (BUSM014)**.
- This course is designed for Master's students, covering basic elements of **statistical methods**, including descriptive statistics, probability, sampling, inference, and multivariate regression analysis. I mainly taught students how to use **Stata** to perform relevant analysis.

Skills

Programming skills: Python, Stata, R, Bash/shell scripting, HPC, SQL

Computational skills: Network science, Causal inference, Bayesian inference, Machine learning, Deep learning, Natural language processing.

Certificates: Neural Networks and Deep Learning
(`https://coursera.org/share/27bcb57c64bc13975dec99a9be379318`;
Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization
(`https://coursera.org/share/d790039cd1ce8b299448df4fe8f1fa5a`)

References

Post-doctoral Research Advisor: Assistant Professor Alexander J. Gates

University of Virginia

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https://alexandergates.net/

Doctoral Research Advisor: Professor Pietro Panzarasa
Queen Mary University of London
p.panzarasa@qmul.ac.uk
http://pietropanzarasa.wordpress.com/

Co-author: Professor Sam Fankhauser
University of Oxford
sam.fankhauser@smithschool.ox.ac.uk
https://www.ox.ac.uk/news-and-events/find-an-expert/professor-sam-fankhauser